

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Chemistry — SSC-I (9th Grade)

Syllabus: Chapter 9 — Chemical Equilibrium

ANSWER KEY (Version 1)

Total Marks: 65

Time Allowed: 2h 30m

Version: 1

SECTION A — Multiple Choice Questions Answer Key (12 × 1 = 12 Marks)

Q1. Which is true about the equilibrium state? ✓ **D — Both reactions continue at the same rate**

1 mark

The textbook defines chemical equilibrium as "**the state of a chemical reaction where the forward and reverse reactions take place at the same rate.**" The reactions do not stop (Key Points, p. 138). Options A, B, and C all imply reactions stopping, which is incorrect. Equilibrium is *dynamic*, not static.

Q2. A reversible reaction is represented by the symbol: ✓ **C — $\rightleftharpoons$ (double arrow)** 1 mark

The textbook states: "**A double arrow $\rightleftharpoons$ in a chemical equation indicates that the reaction is reversible.**" (p. 135). A single arrow ($\rightarrow$) denotes an irreversible/complete reaction. Options B and D show variations that are not the standard double-arrow symbol used in the textbook.

Q3. In an irreversible reaction, equilibrium is: ✓ **C — never established** 1 mark

This is directly from Review Question 1(v) (p. 139). In an irreversible (complete) reaction, reactants are fully converted to products with no back-reaction possible, so equilibrium between forward and reverse reactions is **never established**. Option D is a distractor: the reaction can stop without establishing equilibrium.

Q4. Which does NOT happen when a system is at equilibrium? ✓ **A — Forward and reverse reactions stop** 1 mark

This is directly from Review Question 1(iv) (p. 139). At equilibrium: rates become equal (B is true), concentrations stop changing (C is true), and reaction continues in both directions (D is true). The only false statement is A: reactions do **not** stop at equilibrium.

Q5. Hydrated copper(II) sulphate $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is: ✓ **B — a blue solid** 1 mark

Key Points (p. 138): "**Hydrated copper(II) sulphate is a blue solid.**" Anhydrous CuSO_4 is white. The pink solid belongs to hydrated cobalt(II) chloride, which is a common distractor. This colour difference is testable in Activities.

Q6. When $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is heated, the product formed is: ✓ **A — CuSO_4 (white solid)** 1 mark

The textbook activity states: "When copper(II) sulphate is heated, water is removed, forming **anhydrous copper(II) sulphate, a white solid.**" (p. 137). Options B and C are incorrect: heat does not decompose the sulphate to CuO or Cu at these temperatures.

Q7. Anhydrous cobalt(II) chloride CoCl_2 is: ✓ **B — a blue solid** 1 mark

Key Points (p. 138): "**Anhydrous cobalt(II) chloride is a blue solid.**" Hydrated $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ is pink. Students frequently confuse the hydrated and anhydrous colours of copper sulphate vs. cobalt chloride: for cobalt, anhydrous = blue and hydrated = pink; for copper sulphate, it is the reverse (anhydrous = white, hydrated = blue).

Q8. When water is added to anhydrous CoCl_2 , the colour change is: ✓ **B — blue to pink** 1 mark

The textbook states: "When water is added to anhydrous cobalt(II) chloride (blue), it absorbs water and equilibrium shifts to the left to form hydrated $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ (**pink**)." (p. 137). So the colour change is **blue (anhydrous) to pink (hydrated)**. The reverse (pink to blue) occurs on heating.

Q9. $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$ at equilibrium: substances present are: ✓ **D — H_2 , I_2 , and HI** 1 mark

This is from Review Question 1(ii) (p. 138). At equilibrium, **both reactants and products are always present** because the forward and reverse reactions continue simultaneously. H_2 and I_2 have not fully reacted (the forward reaction is not complete), and HI is decomposing back. Option A would only be correct before the reaction starts.

Q10. Concentrations at equilibrium remain unchanged provided: ✓ **D — all three conditions are maintained** 1 mark

This is from Review Question 1(iii) (p. 138). Section 9.2 lists three conditions that must all be met simultaneously: (1) concentration of reactants or products remains unchanged, (2) temperature is constant, and (3) pressure/volume is constant. Changing any one of these disrupts equilibrium.

Q11. When $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ is heated, equilibrium shifts right because: ✓ **B — Heating favours the endothermic (forward) reaction to minimise temperature increase** 1 mark

The textbook states: "When temperature is increased, equilibrium will shift to favour reactions which will **decrease the temperature**. So, **endothermic reaction is favoured** to minimise the change." (p. 137). Dehydration of $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ (forward direction) is endothermic, so it is favoured on heating. This is Le Chatelier's principle applied to heat.

Q12. Chemical equilibrium is called *dynamic* because: ✓ **C — reactions do not stop; molecules are constantly reacting but concentrations remain constant** 1 mark

The textbook: "**Chemical equilibrium is dynamic equilibrium. This is because reactions do not stop when they reach equilibrium. Individual molecules are constantly reacting.** However, the actual quantities of reactants and products do not change." (p. 136). Concentrations changing (A) would mean equilibrium is not yet reached. Only the forward reaction occurring (B) would describe an irreversible reaction.

SECTION B — Short Questions Answer Key (11 × 3 = 33 Marks)

2(i) — Reversible reaction: definition + two examples 3 marks

Definition: A reaction in which **the products can react together to re-form the original reactants** is called a reversible reaction. A reversible reaction proceeds in both the forward and reverse directions under the same conditions. 1 mark

Symbol: A double arrow ($\rightleftharpoons$) is used to indicate that the reaction is reversible. 1 mark

Two examples (any two from the textbook): 1 mark

(i) $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ (V2O5, 450°C, 200 atm) (ii) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ (Fe, 400°C, 200 atm) (iii) $\text{N}_2(\text{g}) + 2\text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$ (electric spark)

OR

Complete reaction: A reaction in which **all reactants are converted completely into products** and the reaction does not reverse. The reaction goes to completion. 1 mark

Difference: In a complete reaction, products cannot re-form reactants; in a reversible reaction, products react to give back original reactants simultaneously. A single arrow ($\rightarrow$) shows a complete reaction; a double arrow ($\rightleftharpoons$) shows a reversible reaction. 1 mark

Example of complete reaction: $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ (combustion). **Example of reversible reaction:** $\text{N}_{2(\text{g})} + 3\text{H}_{2(\text{g})} \rightleftharpoons 2\text{NH}_{3(\text{g})}$. 1 mark

2(ii) — Chemical equilibrium defined; dynamic nature; two characteristics 3 marks

Definition: "The state of a chemical reaction where the forward and reverse reactions take place at the same rate is called chemical equilibrium." (textbook p. 136) 1 mark

Why dynamic: Equilibrium is dynamic because the reactions do *not* stop at equilibrium. Individual molecules are constantly reacting in both directions, but because the rate of the forward reaction equals the rate of the reverse reaction, the concentrations of all species remain constant. 1 mark

Two characteristics of the equilibrium state: 1 mark

1. The concentrations of reactants and products become constant (do not change over time).
2. Both forward and reverse reactions continue to occur simultaneously at the same rate.

OR

Using $2\text{SO}_{2(g)} + \text{O}_{2(g)} \rightleftharpoons 2\text{SO}_{3(g)}$:

Initial state: Only SO_2 and O_2 are present. The rate of the forward reaction (forming SO_3) is at its highest because reactant concentration is maximum. The rate of the reverse reaction is zero (no SO_3 present yet).

1 mark

As reaction progresses: SO_3 begins to form. The concentration of SO_2 and O_2 decreases (forward rate slows). SO_3 concentration increases and it begins to decompose back (reverse rate increases). 1 mark

Equilibrium: Eventually the rate of the forward reaction equals the rate of the reverse reaction. Concentrations of SO_2 , O_2 , and SO_3 all remain constant. Equilibrium is established. 1 mark

2(iii) — Three conditions for equilibrium 3 marks

From Section 9.2 of the textbook, the three conditions that must be maintained for equilibrium to persist are:

1. The **concentration** of any reactant or product remains unchanged. 1 mark
2. The **temperature** of the system remains constant. 1 mark
3. The **pressure or volume** of the system remains constant. 1 mark

OR

Closed system: A closed system is one in which **no matter can enter or leave** but energy can be exchanged with the surroundings. 1 mark

Why equilibrium requires a closed system: If the container were open, products (especially gases) would escape, preventing the reverse reaction from occurring. Without a reverse reaction, equilibrium can never be established. For example, in $\text{N}_{2(g)} + 3\text{H}_{2(g)} \rightleftharpoons 2\text{NH}_{3(g)}$, if NH_3 escapes, no reverse reaction occurs.

1 mark

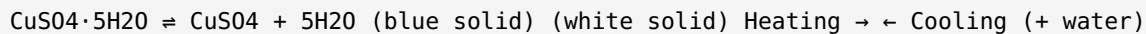
Example: When SO_2 and O_2 gases are mixed in a **sealed container**, SO_3 forms and decomposes back; eventually equilibrium is established. If the container were open, SO_3 vapour would escape and equilibrium would not form. 1 mark

2(iv) — Experiment with $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$: colour changes and equation 3 marks

Colour changes:

1. Initial colour of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$: **blue**. When heated, colour changes from **blue to white** as water is driven off to form anhydrous CuSO_4 . 1 mark
2. When a few drops of water are added to the white anhydrous solid, colour changes from **white back to blue**, confirming the reaction is reversible. 1 mark

Equation:



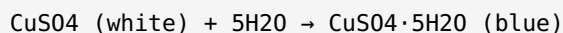
1 mark

OR

Anhydrous CuSO_4 as a test for water: Anhydrous copper(II) sulphate is a **white solid**. When water is added, it absorbs the water and converts back to hydrated $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, which is **blue**. 1 mark

Confirmation: A colour change from **white to blue** confirms the presence of water. This is a simple, visual test for water. 1 mark

Equation:



1 mark

2(v) — Cobalt(II) chloride equilibrium: colours, heating effect 3 marks

Colours (from Key Points, p. 138):

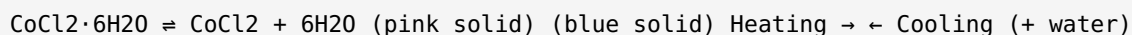
Hydrated $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ is a **pink solid**. 1 mark

Anhydrous CoCl_2 is a **blue solid**. 1 mark

Effect of heating: When the pink hydrated cobalt(II) chloride is heated, it loses its water of crystallisation and becomes anhydrous CoCl_2 (blue). The equilibrium shifts towards the right (towards the anhydrous form). This is because increasing temperature favours the endothermic reaction (dehydration). 1 mark

OR

Equation:



1 mark

Adding water to anhydrous CoCl_2 : When water is added to the **blue** anhydrous CoCl_2 , it absorbs water and the equilibrium shifts to the **left** (reverse direction), forming hydrated $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$. The colour changes from **blue back to pink**. 1 mark

Forward reaction: $\text{CoCl}_2 \cdot 6\text{H}_2\text{O} \rightarrow \text{CoCl}_2 + 6\text{H}_2\text{O}$ (dehydration, on heating).

Reverse reaction: $\text{CoCl}_2 + 6\text{H}_2\text{O} \rightarrow \text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ (rehydration, on adding water). 1 mark

2(vi) — Effect of increasing temperature on equilibrium 3 marks

General principle (Section 9.3): "When temperature is increased, equilibrium will shift to favour the reactions which will **decrease the temperature**. So, endothermic reaction is favoured to minimise the change." 1 mark

Direction of shift: The equilibrium shifts in the direction that **absorbs heat** (the endothermic direction), thereby opposing the temperature increase. 1 mark

Example using $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$: Heating $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ shifts the equilibrium to the right (forward direction): $\text{CuSO}_4 \cdot 5\text{H}_2\text{O} \rightarrow \text{CuSO}_4 + 5\text{H}_2\text{O}$. The dehydration reaction is endothermic (it absorbs heat). The blue colour changes to white. 1 mark

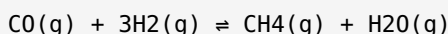
OR

Effect of cooling: When a chemical equilibrium system is cooled (temperature decreased), the system opposes the change by shifting in the **exothermic direction** (the direction that releases heat), thereby raising the temperature back towards the original. 1 mark

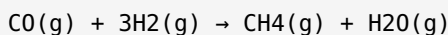
Cobalt(II) chloride equilibrium on cooling: For $\text{CoCl}_2 \cdot 6\text{H}_2\text{O} \rightleftharpoons \text{CoCl}_2 + 6\text{H}_2\text{O}$, on cooling the equilibrium shifts to the **left** (reverse direction). CoCl_2 reabsorbs water to re-form $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$. The colour changes from **blue back to pink**. This is the exothermic (reverse) direction. 2 marks

2(vii) — $\text{CO} + 3\text{H}_2 \rightleftharpoons \text{CH}_4 + \text{H}_2\text{O}$: forward and reverse reactions 3 marks

The full reversible reaction (from textbook exercise Q3, p. 139):

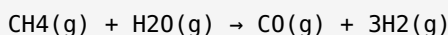


Forward reaction: The reaction in which reactants convert to products (left to right): 1 mark



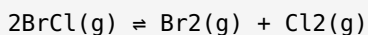
Coal reacts with hot steam to form CO and H_2 . These react with a catalyst to give methane (CH_4) and water vapour. 1 mark

Reverse reaction: The reaction in which products convert back to reactants (right to left): 1 mark



OR

Reversible equation for BrCl decomposition (Think Tank, p. 139):



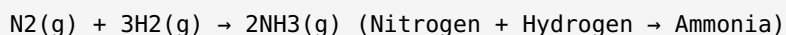
1 mark

Forward reaction: $2\text{BrCl}_{\text{(g)}} \rightarrow \text{Br}_{2\text{(g)}} + \text{Cl}_{2\text{(g)}}$. Bromine chloride decomposes to form bromine and chlorine gases. 1 mark

Conditions favouring reverse reaction: Decreasing temperature (cooling) would favour the exothermic reverse reaction ($\text{Br}_2 + \text{Cl}_2 \rightarrow 2\text{BrCl}$). Also, increasing pressure (or increasing concentration of Br_2 and Cl_2) would shift equilibrium left, favouring formation of BrCl. 1 mark

2(viii) — $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$: forward, reverse, conditions 3 marks

(a) Forward reaction (reactants to products, left to right): 1 mark



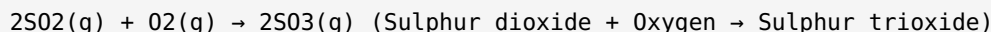
(b) Reverse reaction (products back to reactants, right to left): 1 mark



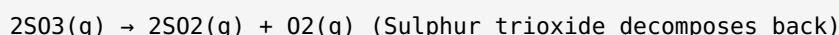
Conditions from textbook (p. 135): Catalyst **Fe (iron)**, temperature **400°C**, pressure **200 atm**. 1 mark

OR

(a) Forward reaction for SO_2/SO_3 equilibrium: 1 mark



(b) Reverse reaction: 1 mark

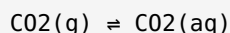


Industrial conditions (textbook p. 135): Catalyst **V_2O_5 (vanadium pentoxide)**, temperature **450°C**, pressure **200 atm**. This is the Contact Process for manufacturing sulphuric acid. 1 mark

2(ix) — Fizzy drinks CO_2 example: physical equilibrium 3 marks

From Science Titbits, p. 136:

Equilibrium equation:



1 mark

When sealed: CO_2 is dissolved in the liquid drink under pressure. The forward reaction ($\text{CO}_{2(\text{g})} \rightarrow \text{CO}_{2(\text{aq})}$) and reverse reaction ($\text{CO}_{2(\text{aq})} \rightarrow \text{CO}_{2(\text{g})}$) occur at the same rate. Equilibrium is maintained. 1 mark

When lid is removed: The pressure above the liquid drops suddenly. The equilibrium shifts to the left (towards the gaseous phase). CO_2 dissolved in the liquid comes out of solution and **bubbles of CO_2 appear**. When the lid is replaced, pressure is restored and bubbles stop: equilibrium re-establishes. 1 mark

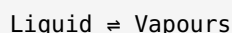
OR

Liquid-vapour equilibrium in a closed container:

Initial state: Liquid is placed in a closed container. Initially the rate of **evaporation** is high (liquid molecules escape into vapour phase) and the rate of **condensation** is zero (no vapour present). 1 mark

As time progresses: Vapour accumulates above the liquid. More vapour molecules collide with the liquid surface and condense. Rate of condensation increases while rate of evaporation decreases (as liquid molecules escape). 1 mark

Equilibrium: Eventually the **rate of evaporation equals the rate of condensation**. The amounts of liquid and vapour remain constant. Equilibrium equation:



1 mark

2(x) — Forward vs reverse reactions; $\text{N}_2 + 2\text{O}_2 \rightleftharpoons 2\text{NO}_2$ example 3 marks

Forward reaction: The reaction in which **reactants convert into products** (proceeds from left to right). It is shown by a forward arrow. 1 mark

Reverse reaction: The reaction in which **products convert back into the original reactants** (proceeds from right to left). It occurs simultaneously with the forward reaction in a reversible system. 1 mark

Example from $\text{N}_2 + 2\text{O}_2 \rightleftharpoons 2\text{NO}_2$:

Forward: $\text{N}_2(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow 2\text{NO}_2(\text{g})$ (Nitrogen + Oxygen $\rightarrow$ Nitrogen dioxide) Reverse: $2\text{NO}_2(\text{g}) \rightarrow \text{N}_2(\text{g}) + 2\text{O}_2(\text{g})$ (Nitrogen dioxide $\rightarrow$ Nitrogen + Oxygen)

1 mark

OR

Using $2\text{NO}_{2(\text{g})} \rightleftharpoons \text{N}_{2(\text{g})} + 2\text{O}_{2(\text{g})}$:

Converts reactants to products: The **forward reaction** (left to right) converts NO_2 to N_2 and O_2 : $2\text{NO}_{2(\text{g})} \rightarrow \text{N}_{2(\text{g})} + 2\text{O}_{2(\text{g})}$. 1 mark

Converts products back to reactants: The **reverse reaction** (right to left) converts N_2 and O_2 back to NO_2 : $\text{N}_{2(\text{g})} + 2\text{O}_{2(\text{g})} \rightarrow 2\text{NO}_{2(\text{g})}$. 1 mark

At equilibrium: The rate of the forward reaction equals the rate of the reverse reaction. Concentrations of NO_2 , N_2 , and O_2 all become constant. Both reactions continue simultaneously at the same rate. 1 mark

2(xi) — Industries using equilibrium / Key Points from Chapter 9 3 marks

From the textbook introduction (p. 135): "Equilibrium reactions are involved in a number of steps in the commercial production of many important chemicals such as **ammonia** and **sulfuric acid**."

Industries and reactions relying on equilibrium: 1 mark each, up to 3 marks

1. **Ammonia (Haber Process):** $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ (Fe catalyst, 400°C , 200 atm)
2. **Sulphuric acid (Contact Process):** $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$ (V_2O_5 , 450°C , 200 atm)
3. **Nitrogen dioxide production:** $\text{N}_2 + 2\text{O}_2 \rightleftharpoons 2\text{NO}_2$ (electric spark)

Importance: Understanding equilibrium allows industry to select conditions (temperature, pressure, catalyst) that maximise yield of the desired product while minimising energy costs.

OR

Key Points from Chapter 9 (p. 138): 1 mark per point, any 4

1. A reaction in which the products can react together to re-form the original reactants is called a **reversible reaction**.
2. A reversible reaction is shown by the double arrow symbol ($\rightleftharpoons$).
3. **Anhydrous copper(II) sulphate** is a **white solid**.
4. **Hydrated copper(II) sulphate** is a **blue solid**.
5. **Anhydrous cobalt(II) chloride** is a **blue solid**.
6. **Hydrated cobalt(II) chloride** is a **pink solid**.
7. A state of a chemical reaction in which forward and reverse reactions take place at the same rate is called **chemical equilibrium**.

3(i) — Reversible reactions, equilibrium definition + three conditions 5 marks

(a) Definitions and distinction: [3 marks]

Reversible reaction: A reaction in which the products can react together to re-form the original reactants. It proceeds in both the forward and reverse directions under the same conditions and **never ends**. Shown by $\rightleftharpoons$. 1 mark

Chemical equilibrium: The state of a chemical reaction where the forward and reverse reactions take place at the **same rate**. Concentrations of reactants and products remain constant. 1 mark

Feature	Complete (Irreversible) Reaction	Reversible Reaction
Direction	One direction only ($\rightarrow$)	Both directions ($\rightleftharpoons$)
Completion	All reactants convert to products	Mixture of reactants and products at equilibrium
Equilibrium	Never established	Established in a closed system
Example	$2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$	$\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$

1 mark

(b) Three conditions for equilibrium: [2 marks]

1. Concentration of any reactant or product remains unchanged. $\frac{2}{3}$ mark
2. Temperature of the system remains constant. $\frac{2}{3}$ mark
3. Pressure or volume of the system remains constant. $\frac{2}{3}$ mark

Effect if one changes: If any condition is altered, the equilibrium is disturbed and the system shifts in the direction that opposes the change (to minimise the disturbance). For example, increasing temperature shifts equilibrium towards the endothermic direction. 1 mark

OR

(a) Equilibrium established step by step in $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$: [3 marks]

Step 1 (Initial state): Only SO_2 and O_2 are present in the sealed container. Rate of forward reaction is maximum; rate of reverse reaction = 0. No SO_3 has formed yet. 1 mark

Step 2 (Reaction progresses): SO_3 begins to form (forward reaction). Concentration of SO_2 and O_2 decreases, slowing the forward rate. SO_3 begins to decompose (reverse reaction starts). Reverse rate increases. 1 mark

Step 3 (Equilibrium): Forward rate = reverse rate. Concentrations of SO_2 , O_2 , and SO_3 all become constant. Equilibrium is established. The reaction continues in both directions but with no net change in concentrations. 1 mark

(b) Dynamic vs static equilibrium: [2 marks]

Dynamic equilibrium: Both forward and reverse reactions continue to occur; molecules are constantly reacting, but concentrations remain constant because both rates are equal. 1 mark

Static equilibrium: No reactions are occurring; everything has stopped. Chemical equilibrium is NOT static.

Analogy: Imagine escalators in a busy mall: if exactly as many people step on at the bottom as step off at the top each second, the number of people on the escalator stays constant, even though everyone is moving. The system is dynamic. 1 mark

3(ii) — $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ experiment + equation with labels 5 marks

(a) Laboratory experiment on $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$: [3 marks]

Materials required (from textbook p. 137): Test tube, dropper, heating system, hydrated copper(II) sulphate. ½ mark

Procedure:

1. Place 5 g of hydrated copper(II) sulphate (blue) in a test tube and heat slowly. ½ mark
2. Observe the colour change from **blue to white** as water is driven off. ½ mark
3. Allow the test tube and contents to cool to room temperature.
4. Add a few drops of water using a dropper. Observe the colour change from **white back to blue**. ½ mark

Observations:

- On heating: colour changes **blue** → **white** (hydrated to anhydrous). Water is released as vapour.
- On adding water: colour changes **white** → **blue** (anhydrous to hydrated). Heat may be felt (exothermic).

1 mark

Conclusion: The reaction is **reversible**. Heat drives the forward reaction; adding water drives the reverse reaction. ½ mark

(b) Reversible equation with labels: [2 marks]

Heating → $\text{CuSO}_4 \cdot 5\text{H}_2\text{O} \rightleftharpoons \text{CuSO}_4 + 5\text{H}_2\text{O}$ (blue solid) (white solid) ← Cooling / adding water

1 mark (equation correct) + 1 mark (colours and direction labels correct)

OR

(a) Cobalt(II) chloride equilibrium: [3 marks]

Colours: $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ (hydrated) is **pink**; CoCl_2 (anhydrous) is **blue**. 1 mark

(i) Effect of heating: When the pink hydrated solid is heated, water is removed. The equilibrium $\text{CoCl}_2 \cdot 6\text{H}_2\text{O} \rightleftharpoons \text{CoCl}_2 + 6\text{H}_2\text{O}$ shifts to the **right**. Colour changes from **pink to blue**. The forward (dehydration) reaction is endothermic. 1 mark

(ii) Effect of adding water: When water is added to the blue anhydrous CoCl_2 , it absorbs water and equilibrium shifts to the **left**. $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ reforms. Colour changes from **blue back to pink**. 1 mark

(b) Equation and direction of shift on heating: [2 marks]

Heating → $\text{CoCl}_2 \cdot 6\text{H}_2\text{O} \rightleftharpoons \text{CoCl}_2 + 6\text{H}_2\text{O}$ (pink solid) (blue solid) ← Cooling / adding water

On heating, equilibrium shifts **right** (endothermic direction). The textbook principle: "When temperature is increased, equilibrium shifts to favour reactions which decrease the temperature (endothermic reaction)."

2 marks

3(iii) — Temperature effects on equilibrium + effect of adding water 5 marks

(a) General principle and predictions: [3 marks]

General principle (Section 9.3): When temperature is increased, the equilibrium shifts to favour the **endothermic reaction** (the reaction that absorbs heat), thus opposing the temperature increase. 1 mark

For $\text{CoCl}_2 \cdot 6\text{H}_2\text{O} \rightleftharpoons \text{CoCl}_2 + 6\text{H}_2\text{O}$:

The forward reaction (dehydration) is endothermic (requires heat).

The reverse reaction (hydration) is exothermic (releases heat).

(i) On heating: Equilibrium shifts to the **right** (forward direction). More CoCl_2 (blue) and H_2O are formed. Colour changes from **pink to blue**. 1 mark

(ii) On cooling: Equilibrium shifts to the **left** (reverse direction). More $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ (pink) is formed. Colour changes from **blue back to pink**. The exothermic reverse reaction is favoured. 1 mark

(b) Effect of adding water (increasing concentration): [2 marks]

When water is added to the cobalt(II) chloride equilibrium system, the concentration of H_2O (a product) increases. The system opposes this change by shifting the equilibrium to the **left** (reverse direction), consuming the excess water and forming more $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ (pink). 1 mark

This demonstrates that changing the concentration of a product shifts equilibrium towards the reactant side. The colour change from **blue to pink** confirms the leftward shift. 1 mark

OR

(a) $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$: forward, reverse, and conditions: [3 marks]

Forward reaction:

$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$ (Nitrogen reacts with hydrogen to form ammonia)

1 mark

Reverse reaction:

$2\text{NH}_3(\text{g}) \rightarrow \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$ (Ammonia decomposes back to nitrogen and hydrogen)

1 mark

Conditions: Catalyst: **Fe (iron)**; Temperature: **400°C**; Pressure: **200 atm**. These conditions are chosen industrially to obtain a reasonable yield of ammonia at an acceptable rate. 1 mark

(b) Closed system and equilibrium: [2 marks]

Closed system: A system in which no matter enters or leaves (but energy exchange is possible). 1 mark

Open system disrupts equilibrium: In an open system, gaseous products escape into the atmosphere. The reverse reaction cannot occur because products are no longer present. Equilibrium is never established. For example, if NH_3 escaped from the reaction vessel in the Haber Process, N_2 and H_2 would keep reacting forward without a reverse reaction. 1 mark

3(iv) — Key Points summary + reversible vs irreversible comparison table **5 marks****(a) Key Points of Chapter 9 (from p. 138): [3 marks]**

1. A reaction in which the products can react together to re-form the original reactants is called a **reversible reaction**. $\frac{1}{2}$
2. A reversible reaction is shown by the symbol $\rightleftharpoons$ (double arrow). $\frac{1}{2}$
3. **Anhydrous copper(II) sulphate** is a **white solid**. $\frac{1}{2}$
4. **Hydrated copper(II) sulphate** is a **blue solid**. $\frac{1}{2}$
5. **Anhydrous cobalt(II) chloride** is a **blue solid**. $\frac{1}{2}$
6. **Hydrated cobalt(II) chloride** is a **pink solid**. $\frac{1}{2}$
7. A state of a chemical reaction in which forward and reverse reactions take place at the same rate is called **chemical equilibrium**.

1 mark (7th point)**(b) Comparison table: [2 marks]**

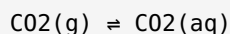
Heading	Reversible Reaction	Irreversible (Complete) Reaction
(i) Direction	Both forward and reverse simultaneously	Forward direction only
(ii) Equilibrium	Established in a closed system	Never established
(iii) Example	$\text{CuSO}_4 \cdot 5\text{H}_2\text{O} \rightleftharpoons \text{CuSO}_4 + 5\text{H}_2\text{O}$	$2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ (combustion)
(iv) Symbol	$\rightleftharpoons$ (double arrow)	$\rightarrow$ (single arrow)

2 marks**OR****(a) Liquid-vapour equilibrium (Liquid $\rightleftharpoons$ Vapours): [3 marks]**

Start: Liquid is placed in a **closed container**. Some liquid evaporates: faster molecules escape from the surface into the vapour phase. Rate of evaporation is at its highest; rate of condensation = 0 (no vapour yet). **1 mark**

Progress: As more liquid evaporates, vapour density above the liquid increases. Vapour molecules begin to collide with the liquid surface and condense back. Rate of condensation increases; rate of evaporation decreases (fewer energetic molecules remaining). **1 mark**

Equilibrium: Rate of evaporation = rate of condensation. The amount of liquid and the amount of vapour both remain constant. This is **physical equilibrium**. Even though individual molecules continue to evaporate and condense, the macroscopic amounts do not change. **1 mark**

(b) CO₂ fizzy drinks and pressure effect: [2 marks]**Equilibrium equation:**

(i) Lid sealed: CO₂ is under high pressure inside the bottle. The equilibrium favours the dissolved state (right): CO₂(aq) is abundant; only a little CO₂(g) is present. No bubbles form. **1 mark**

(ii) Lid removed: Pressure above the liquid drops to atmospheric pressure. The equilibrium shifts to the **left** (towards gaseous CO₂): dissolved CO₂ comes out of solution and **bubbles appear**. Replacing the lid restores pressure and stops bubbling (equilibrium re-establishes). **1 mark**